

MUHAMMAD FAIZAN

AI Software Engineer | Machine Learning Engineer

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SUMMARY

Passionate AI/ML engineer and full-stack developer with hands-on experience in machine learning, NLP, computer vision, agentic AI and generative AI. Skilled at building AI-powered systems, web applications. Experienced with building RAG, AI agents, vector databases.

EDUCATION

Bachelor of Science in Computer Science (BSCS)-3.47

November 2021 - June 2025

Namal University, Pakistan

TECHNICAL SKILLS

Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, Generative AI, Agentic AI, Applied AI, Natural Language Processing (NLP), BERT, Embeddings, Semantic Similarity, Speech-to-Text, Text-to-Speech, Retrieval-Augmented Generation (RAG), AI Agents, Workflow Automation, Vector Databases, Pgvector, Computer Vision, Image Processing, Object Detection, Face Verification, CNNs, RNNs, OpenCV, DeepFace, TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, Python, SQL, Full-Stack Development, MERN, Node.js, React, PostgreSQL, Git, Data Science, Data Processing, Data Augmentation, Agile Methodology, LangChain

PROFESSIONAL EXPERIENCE

AI Software Engineer

Next Generation Technology Solutions (NGTSOL)

June 2025 – Present

- AI Engineer specializing in building production-grade intelligent systems combining Machine Learning, Generative AI (LLMs), agentic AI, and full-stack engineering to automate enterprise workflows and decision-making.
 - Built an **AI SOC Analyst** platform with **multi-tool AI agent** integrating Wazuh SIEM data, automated health monitoring, threat intelligence enrichment, LLM-based alert summarization, and investigation plan generation to assist cybersecurity analysts with faster triage and decision support.
 - Architected and delivered an **AI-Powered Recruitment Platform** (PERN Stack) integrating semantic resume parsing, intelligent candidate-job matching, job posting and approval workflow, and recruitment automation, significantly improving hiring efficiency and reducing manual screening effort.
 - Developed a **Real-Time Incident Classification & Auto-Routing System** by training a custom machine learning model to categorize ServiceNow incidents and assign them to appropriate teams with 83% accuracy, accelerating response times and reducing operational overhead.
 - Built and integrated AI capabilities including chatbots, semantic search, NLP embeddings, and workflow automation into enterprise applications.
 - Full-stack development using React.js, Node.js, Express.js, PostgreSQL, REST APIs, ensuring scalable architecture, optimized AI inference performance, and secure system design.
 - Collaborated with cross-functional teams and clients to translate business requirements into production systems using Agile methodologies and best engineering practices.
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FINAL YEAR PROJECT (Award Winning (1 st Prize) at Bahria University AI Innofest 2025)	
AI-Powered Interviewing System	Sep 2024-April 2025
Built an AI-driven interview automation platform designed to conduct job interviews with AI Integration. The system acts as a virtual interviewer, generates dynamic questions using LLMs and evaluates candidate responses in real time, including candidates' confidence prediction with facial expressions and cheating detections.	
Key Highlights of the FYP:	
• Automated Interviews: Conducts real-time, adaptive interviews without human intervention. • Smart Questioning: Uses LLMs to generate context-aware, tailored questions based on job requirements. • AI-Powered Evaluation: Performs speech-to-text conversion, semantic similarity checks, and intent analysis to assess technical and soft skills. • Face Verification & Cheating Detection: Integrates DeepFace + MTCNN to ensure candidate authenticity. • Confidence Prediction: Uses deep learning to predict candidate confidence levels dynamically. • Impact: Reduced manual screening time by ~70% and improved shortlisting efficiency for hiring teams.	
OTHER PROJECTS	
Urdu Ligature Recognition with deep learning	May-June 2024
• Built a neural network using AlexNet, VGG16, and BLSTM for Urdu ligature recognition, achieving 83% accuracy.	
Real-Time Magic Spell Classifier (2 nd prize at LUMS AI Nexus Hackathon 2024-25)	December 2024
• Created software for real-time data collection of wand movements, compiling a custom dataset with 25 spells and 100 frames per spell for classification.	
Networking Tool Development	Jan-Feb 2024
• Created an interactive tool for designing and configuring network topologies with validation and simulation features.	
Social Media Platform:	June-July 2022
• Built a platform with posting, liking, and connection features from scratch in python	
AI Based Pacman Game	March 2024
• Implemented an AI-driven Pacman game using Pygame, with SQLite for storing game data and high scores.	
BERT Fine Tuning	August 2024
• Fine-tuned BERT on customized (self collected data) for high-accuracy intent classification in NLP tasks.	
Examination Cheating Detection System	September 2024
• Developed an innovative system for predicting cheating in record-based exams, with machine learning techniques to detect unusual patterns in grades.	

FREELANCE EXPERIENCE

Mar 2023- Jan 2025

- **Python & ML Tutor:** Conducted more than 50 online sessions on Python and machine learning fundamentals, helping students around the world with hands-on coding and projects.
 - **Software Debugging & Design:** Assisted clients with software debugging, design patterns, and requirement specifications.
 - **Data Science & ML Consultant:** Worked on data analysis and machine learning projects, offering solutions in models training, and evaluation.
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RESEARCH WRITINGS

- **Role of AI in Cyber Attack Predictions and Prevention:** Led a research initiative on utilizing AI models for predicting and preventing cyber-attacks, contributing to the development of proactive security systems through machine learning-based threat detection.
 - **Enhance OCR Capabilities with Bi-directional Long-Short Term Memory:** Conducted research focused on improving Optical Character Recognition (OCR) accuracy, exploring advanced techniques to increase recognition efficiency and robustness in various contexts.
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ACHIEVEMENTS & AWARDS

- **Best Final Year Project – 1st Prize** **May 2025**
Won 1st place at Bahria AI Innofest for developing an AI-Powered Interviewing System, competing against top universities nationwide.
- **2nd Prize – LUMS AI Nexus Hackathon** **Apr 2025**
Secured 2nd position among teams from across Pakistan by building an AI-driven solution under 48 hours.
- **2nd Prize – Namal University Programming Hackathon** **Feb 2024**
Achieved 2nd place in an inter-university programming competition, solving complex algorithmic challenges.